Security bulletin TN1489-ST-PSIRT: Physical attacks on STM32 and STM32Cube firmware

Overview

This security bulletin relates generally to any physical attack against any version of an STM32 and/or STM32Cube firmware product (collectively referred to as ***STM32 product*** in this document). It does not pertain to any specific known physical attack.

Physical attacks mean attacks made by acquiring physical access, or very close access, to an *STM32 product*. This includes, but may not be limited to:

- attacks made through access to physical interfaces
- perturbation attacks (that is, fault injections to induce an exploitable error)
- side channel attacks (including timing attacks), power analysis (SPA, DPA, and the like), and electromagnetic analysis
- invasive attacks including physical changes and reverse engineering

In addition to requiring physical or close access to an *STM32 product*, *physical attacks* often require specialized tools and techniques.

Description

Regarding *STM32 products* and their resistance to *physical attacks*:

Unless an *STM32 product* is SESIP or PSA certified as having a security assurance level covering ***physical attacker resistance***, it may be vulnerable to *physical attacks*.

For the security assurance level of a certified *STM32 product*:

- Services and features that are protected against *physical attacks* are identified in the certification scope of the product ***security targets***. Note that the protection level (security assurance level) depends on the certification level obtained.
- Resistance level for those services and features is described in the *security targets*.
- Services and features outside the certification scope are not covered.

If a service or a feature of an *STM32 product* is not certified as having *physical attacker resistance*, such *STM32 product* should not be considered resistant to *physical attacks*.

Generally speaking, because *physical attacks* require physical or close access to an *STM32 product*, they are typically limited to only a select number of devices to which attackers have physical access, as opposed to remote attacks. The potential impact of a *physical attack* is therefore typically much more limited than the impact of a remote attack. The overall impact of a *physical attack* also highly depends on the sensitivity of the information associated with the targeted product.

When the application context requires resistance to *physical attacks*, it is advised to select *STM32 products* that are SESIP or PSA certified with *physical attacker resistance*. To meet the application requirements, it may not be sufficient to select a SESIP or PSA certified *STM32 product*. It is also important that the security assurance level stated in that certification provides *physical attacker resistance*.

For certification scope and security assurance level, refer to the product *security targets* and certificates available on the certification body websites:

- https://www.trustcb.com/iot/sesip/sesip-certificates/
- https://www.psacertified.org/certified-products/

Contact information

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Revision history

Table 1. **Document revision history**

Date	Version	Changes
09-Oct-2023	1	Initial version.

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